

Q23 Average e.m.f. induced,

$$\begin{aligned} E &= \left| -\frac{\Delta\Phi}{\Delta t} \right| \\ &= \frac{\Phi_f - \Phi_i}{\Delta t} \\ &= \frac{NBA \cos 0 - NBA \cos 45^\circ}{\Delta t} \\ &= \frac{3000 \times 1.8 \times \frac{\pi}{4} (0.020)^2}{0.060} (1 - \cos 45^\circ) \\ &= 8.3 \text{ V} \end{aligned}$$

Answer: A

Q24 When the current is flowing from X to Y the effective resistance of the circuit is $2.0 \, \Omega$